CIVICSITE AI

Binghamton housing site suitability

Data engineering and GIS field guide for a tract-level, transparent hackathon prototype

ONE DELIVERABLE. TWO PARTS.

Part I - exact data-person work plan Part II - end-to-end build guide

Scope: Broome County census tracts, with Binghamton highlighted Decision support, not automated siting approval

Prepared for a rapid civic-tech demo. Version 1.0 | November 2026

Start here

Front-page checklist

The minimum viable product is a map and a small table where every census tract has a reproducible score, explainable inputs, and visible limitations. Build that before adding AI polish.

Goal: identify tracts that combine high housing need, usable transit/jobs access, and lower flood exposure. This is a screening and prioritization tool, not a claim that a tract is buildable, profitable, or approved.

The data person must complete

- ■ Create one canonical tract file keyed by 11-digit GEOID (string, leading zeros retained).
- ■ Load TIGER/Line tract polygons; verify Broome County FIPS is 007 and New York state FIPS is 36.
- ■ Create transit access at tract level using stop/route coverage or line-buffer coverage; state the method.
- ■ Overlay FEMA flood hazard polygons; calculate tract area share in mapped flood zones in a projected CRS.
- ■ Export final GeoJSON and CSV; run row-count, join, range, and map sanity checks.
- ■ Download and document ACS 5-year indicators; keep estimates and margins of error when available.
- ■ Convert GTFS shapes.txt from points to lines in QGIS using shape_id and shape_pt_sequence.
- ■ Aggregate LODES workplace jobs (WAC) or an explicit proxy to tract geography; do not call it travel-time access unless calculated.
- ■ Normalize raw measures to 0-100 scores, preserve the raw values, and write scoring_config.md.
- ■ Hand off source URLs, dates, scripts/QGIS models, data dictionary, and known limitations.

Suggested 24-hour sequence

Hours	Data/GIS owner	App/story owner	Gate
0-2	Set scope, folder structure, GEOID contract; fetch boundaries and ACS.	Frame user story, rough UI, define what the score means.	Shared rubric accepted.
2-6	Clean ACS; import GTFS; make first tract joins.	Build upload/read path and score explanation panel.	First map with no broken joins.
6-12	Add jobs and flood indicator; calculate raw metrics.	Connect GeoJSON; show ranked tracts and caveats.	MVP score runs end-to-end.
12-18	QA, edge cases, sensitivity test; export final files.	Polish UI; write demo script and fallback screenshots.	Demo-ready candidate.
18-24	Freeze data, package provenance, rehearse handoff.	Rehearse 3-minute narrative and Q&A.;	One reproducible release folder.

Project frame

What you are building

CivicSite AI is an explainable tract-level housing opportunity screen for Binghamton and Broome County. It helps a civic user ask: *where do housing need, access, and lower environmental exposure align?* It does not replace parcel analysis, engineering review, zoning review, community engagement, or fair-housing analysis.

Layer of question	What this prototype can answer	What it cannot answer
Housing need	Where households show higher rent burden / poverty / vacancy pressure, depending on selected measures.	Whether a particular household should move or whether a project is financially feasible.
Development suitability	Where tract-level access and lower mapped flood exposure are favorable.	Parcel ownership, slope, utilities, contamination, zoning, capacity, or construction cost.
Public benefit	Where a transparent public-interest rubric indicates potential value.	Private profit, investor return, entitlement approval, or displacement risk without additional work.
AI role	Explain scores, surface provenance, answer guided questions from provided data.	Invent missing facts, make approvals, or hide assumptions.

Use labels such as **screening score**, **tract priority**, or **prototype suitability**; avoid calling results “optimal sites.”

Contents

Part	Use it for
I. Data-person plan	Exact milestones, dataset-by-dataset work, acceptance tests, and handoff.
II. Full build guide	Raw sources through scoring, GIS, app, testing, demo, and final package.
Appendices	Data dictionary, formulas, QA script, repo layout, source links, and QGIS GTFS steps.

Ownership checklist

Part I - exact data-person work plan

Treat this as the exact implementation backlog. Mark a milestone complete only when its acceptance test passes and its output is committed or saved in the shared project folder.

Milestone 0 - lock the data contract

■ Write **scope.md**: study area = all Broome County census tracts; Binghamton can be filtered or highlighted. State the ACS and LODES vintages.

■ Choose one analysis CRS for area and distance. A local projected CRS is required for buffers and area; retain EPSG:4326 only for web export.

■ Create **data/processed/tract_master.csv** with one row per GEOID and the required columns listed in the data dictionary.

■ Store GEOID as text only: state(2)+county(3)+tract(6), e.g. 36007000100. Never convert to an integer.

■ Create **provenance.csv** with source, URL, vintage, accessed date, geographic unit, transformations, and owner.

■ Record the baseline formula and weights in **scoring_config.md**; no unexplained spreadsheet formulas.

Acceptance test

Exactly one row per expected Broome County tract; no null GEOIDs; no duplicate GEOIDs; every input column identifies its vintage and units.

Milestone 1 - boundaries and ACS housing need

■ Download Census TIGER/Line census tracts for New York and select COUNTYFP = 007. Keep original geometry in raw/.

■ Download a consistent ACS 5-year vintage at tract level. Use 5-year, not 1-year, because tract coverage is required.

■ Pull only documented variables. Preserve estimate fields; retrieve margin-of-error fields when time permits.

■ Clean numeric fields, map Census missing/suppression values to null, and preserve the original response file.

■ Calculate a documented housing-need measure, e.g. renter households paying 35% or more of income for gross rent divided by renter households with rent as a percent of income computed.

■ Join ACS to tracts on GEOID; inspect 10 sampled labels/values in the map and confirm no join has multiplied rows.

Acceptance test

The tract layer shows one polygon per GEOID; 100% of expected tracts have an ACS join status of matched or an explicit, documented null.

Execution

Data-person milestones, continued

Milestone 2 - transit access

■ Unzip/retain the full GTFS feed if available. shapes.txt describes alignments; stops.txt, trips.txt, routes.txt, and stop_times.txt are needed for stronger service measures.

■ Validate shapes.txt: required columns exist, lat/lon are numeric and plausible, no duplicate shape_id + shape_pt_sequence keys, sequence increases within shape_id.

■ In QGIS import as EPSG:4326 points, then run Points to Path with group = shape_id and order = shape_pt_sequence.

■ Reproject lines and tracts to the selected local projected CRS before buffering.

■ For a simple MVP, buffer lines (or preferably stops) by a clearly labeled walking-distance proxy, intersect with tracts, and calculate covered area share. Do not equate route geometry with frequency.

■ If stop_times/calendar are available, create a stronger score: count weekday departures at stops within the buffer. Document service date and time window.

Acceptance test

The transit lines appear over Binghamton roads rather than in the ocean or another country; each tract has a raw coverage or service measure and an explicit missing-data flag.

Milestone 3 - jobs access

■ Download New York LODES WAC (workplace area characteristics) for a stated year; this is a block-level jobs-at-workplace dataset.

■ Use the appropriate total-jobs field (commonly C000 in WAC) and the official block crosswalk / geometry to aggregate to tract, or use an explicitly documented spatial join.

■ Label the MVP metric accurately: **jobs located in/near tract** is not the same as travel-time access to jobs.

■ Optional better measure: calculate jobs within a specified buffer/network travel time from tract centroids. Declare distance/network and destination jobs vintage.

■ Check for disclosure-noise / suppression caveats and do not compare LODES jobs directly with ACS population counts.

Acceptance test

Sum of tract jobs is plausible compared with the selected county area; zeros and outliers have been inspected on a map and in a sorted table.

Milestone 4 - flood exposure and optional infrastructure

■ Acquire FEMA NFHL / Map Service Center flood hazard polygons for the study area. Preserve service/download metadata and retrieval date.

■ Reproject flood polygons and tracts to the local projected CRS. Repair invalid geometry only in a derived copy, logging the repair.

■ Intersect flood polygons with tracts and calculate **flood_area_share = mapped flood-zone area / tract land area**. Make a documented decision about water polygons and zone classes.

■ For infrastructure readiness, only add a field if a local authoritative layer exists. Otherwise leave it null and exclude it from the denominator - do not fabricate a proxy.

■ Create map snapshots of flood overlays for at least three tracts: high, medium, low.

Close the data work

Scoring, QA, and handoff

Milestone 5 - calculate, validate, export

■ Keep raw metrics, normalized scores, weights, direction, and final score in the same final table. Never overwrite a raw column with a score.

■ Normalize using county-wide observed values or stated policy thresholds. Clamp outputs to 0-100 and retain the normalization method.

■ Reverse flood exposure so less exposure earns a higher suitability component. Do not reverse housing need: higher need means greater public-benefit priority.

■ Run the QA checklist in Part II. Perform a sensitivity test with at least one alternative weight set.

■ Export **tract_scores.geojson** in EPSG:4326 and **tract_scores.csv**. Validate GeoJSON in the app and QGIS.

■ Hand over an evidence packet: processed files, sources, scoring config, screenshot map, and 5-line explanation of each limitation.

Minimum handoff package

File	Purpose	Must contain
tract_scores.geojson	App map layer	GEOID, names, raw metrics, scores, final_score, data flags.
tract_scores.csv	Inspectable tabular output	Same score fields, no geometry dependency.
data_dictionary.md	Interpretation	Units, direction, source, vintage, transformations.
provenance.csv	Reproducibility	URLs, accessed date, version/vintage, owner.
scoring_config.md	Transparency	Formula, weights, null policy, classification.
qa_report.md	Confidence	Checks run, counts, anomalies, decisions.

Definition guardrail: high housing need is a public-benefit signal; it is not a claim that development is easiest or most profitable. High scores are prompts for further investigation, not approvals.

From raw data to map

Part II - full project guide

1. Set up a small, reproducible pipeline

Freeze a study-area contract before downloading anything. The common failure in a hackathon map is mixing years, geographies, or definitions without noticing. Use one processing notebook/script or an ordered QGIS model so another teammate can rebuild the result.

Source	Role in prototype	Preferred unit	MVP output	Important caution
ACS 5-year	Housing need	Census tract	Rent burden / poverty / vacancy inputs	Survey estimates have margins of error; denominators matter.
TIGER/Line	Boundary geometry	Census tract	Base polygons and GEOID	Use a matching Census vintage when feasible.
GTFS	Transit access	Stops/routes then tract	Buffer/service coverage score	Shapes alone say route path, not service frequency.
LEHD-LODES	Jobs access	Census block -> tract	Workplace jobs or reachable jobs	A job is not a person; vintage and geography must be stated.
FEMA NFHL	Flood exposure	Hazard polygon -> tract	Mapped flood-area share	Coverage and zones are not a full climate-risk study.
Local utility / zoning	Optional readiness	Parcel/area	Separate flag, not invented score	Only use authoritative, current local data.

2. Suggested raw-download log

Field	Example
dataset_id	acs_2024_5year_tract_broome
source_url	Exact API query or download landing page
vintage	2024 ACS 5-year; LODES year; FEMA retrieval date
geography	NY state 36, Broome County 007, tract
downloaded_at	Local date and time
transformations	Filter to county; join key; aggregation; buffers
owner / reviewer	Name or team role

Practical GIS steps

QGIS guide - turn GTFS shapes.txt into routes

Your shapes.txt has the core fields needed to reconstruct linework: **shape_id** identifies the shape, **shape_pt_lat** is Y, **shape_pt_lon** is X, and **shape_pt_sequence** controls the drawing order. It is a point sequence, not a ready-made line layer.

A. Import the points

QGIS setting	Choose
Menu	Layer -> Add Layer -> Add Delimited Text Layer
File	shapes.txt
Geometry definition	Point coordinates
X field	shape_pt_lon
Y field	shape_pt_lat
Geometry CRS	EPSG:4326 - WGS 84
Check before Add	Preview columns; numeric lon/lat; comma delimiter

B. Build route lines

- Open Processing -> Toolbox and search **Points to Path**.
- Input layer: the imported shapes point layer.
- Order expression: **shape_pt_sequence**.
- Group expression: **shape_id**.
- Save output as data/processed/gtfs_shapes_lines.gpkg. Do not overwrite raw shapes.txt.
- Open the attribute table and verify one output feature per shape_id (unless known duplicate IDs were intentionally split).

C. Validate before using for a score

Check	How	If it fails
Coordinate swap	Layer should land around Binghamton; longitude is X and latitude is Y.	Re-import; do not use latitude as X.
Sequence ordering	Inspect a few shapes with arrows/labels or select a single shape.	Confirm numeric type; sort within each shape_id.
Geometry quality	Visual overlay on basemap; look for long diagonals or spikes.	Find duplicated/bad points and document removal.
Route meaning	Join trips.txt -> routes.txt only if you need route labels.	Do not infer route frequency from shape count.
Distance/area	Right-click -> Export -> Save Features As; reproject to local projected CRS.	Do not buffer in degrees for a walking-distance claim.

Recommended transit MVP: If you only have shapes.txt, label this route proximity coverage. If you also have stops.txt plus stop_times.txt/calendar.txt, measure scheduled weekday service near each tract and call it transit service access.

Method design

Build indicators that mean what they say

Use a simple scoring scheme you can explain in 20 seconds. The raw metric, direction, unit, source year, and transformation belong in the output. The suggested components below are a starting point, not a universal policy decision.

Component	Raw measure example	Higher raw value means	Score direction	Notes
Housing need	Share of renter households paying >=35% of income for gross rent; optionally poverty rate / vacancy context.	Greater housing stress	Higher priority	Use correct ACS denominator. Avoid a single measure if it is unstable.
Transit access	Share of tract area within buffer of stops/lines; or weekday departures near residents.	More transit availability	Higher suitability	A buffer is a proxy; frequency is better when GTFS schedules exist.
Jobs access	WAC jobs in/near tract, or jobs reachable under declared travel rule.	More job opportunity nearby	Higher suitability	Call it workplace-job proximity unless you compute travel time.
Flood exposure	Share of tract area in selected FEMA mapped flood zone(s).	More mapped exposure	Lower suitability	Reverse after normalization; state zone and denominator.
Infrastructure readiness	Authoritative utility/sewer/road capacity flag, if available.	More readiness	Higher suitability	Optional. Missing means unknown, not poor.

Normalization

For a positive indicator x (need, transit, jobs), use: **score = 100 x (x - min) / (max - min)**. For a risk indicator such as flood share, use: **score = 100 x (1 - (x - min) / (max - min))**. Clamp to [0, 100]. If max equals min, set the score to null and report that the measure has no variation.

Baseline score

Screening score = 0.35(Housing Need) + 0.25(Transit Access) + 0.25(Jobs Access) + 0.15(Low Flood Exposure). Keep the coefficients editable in one configuration file. If infrastructure is included, rebalance weights explicitly; do not silently add it.

Classification

For a demo, show continuous score and a simple label: 75-100 Higher priority for follow-up; 50-74 Investigate; 0-49 Lower priority under this rubric; null Insufficient data. Never imply precision that the input data do not support.

Repeatable pipeline

Data engineering sequence

Recommended operations, in order

Step	Action	Output / check
1. Ingest	Save untouched downloads in data/raw/ and record checksums or retrieval dates.	Raw file exists; provenance entry created.
2. Standardize keys	Create GEOID from state/county/tract fields; left-pad and keep text.	11-character GEOID only.
3. Standardize CRS	Maintain raw EPSG:4326 where supplied; create projected analysis copies.	All overlays and buffers use same projected CRS.
4. Filter scope	Filter to Broome tracts and selected time/vintage.	Expected tract count documented.
5. Aggregate	Convert point/line/polygon/block data to tract-level raw measures.	One calculated value per GEOID.
6. Join	Left join indicators into canonical tract frame, checking cardinality.	No row multiplication.
7. Score	Normalize, reverse risks, combine weights, flag nulls.	Raw and scored columns coexist.
8. Export	Write CSV and Web GeoJSON; retain GeoPackage for QGIS.	Same GEOID set across final outputs.

Join rules that prevent silent errors

- Use a **left join from tract_master**. This preserves the geography and makes missing inputs visible.
- Assert indicator GEOID is unique before joining. If an indicator has repeated GEOIDs, aggregate first.
- After every join, compare row count before and after; it should not change.
- Keep a source-specific status field such as transit_status = matched / missing / out_of_scope.
- Use numeric types for values, strings for IDs, and ISO dates for retrieval dates.

Illustrative pandas validation

```
assert tract_master['GEOID'].str.fullmatch(r'\d{11}').all()
assert not tract_master['GEOID'].duplicated().any()
assert before_rows == after_rows
assert final['final_score'].dropna().between(0, 100).all()
```

Do not skip

QA protocol and caveats

Run these checks before the app receives data

- Row count: one row and one polygon per tract GEOID; compare expected count against Census boundary source.
- Key check: every GEOID is exactly 11 numeric characters; leading zeros survive CSV re-open.
- Join check: post-join tract count equals pre-join tract count; all unmatched records are reported.
- Range check: percentages are 0-100; area shares are 0-1 or 0-100 consistently; scores are 0-100.
- Geometry check: valid polygons; no empty geometry; layers align visually after reprojection.
- Transit check: GTFS line locations look plausible; sample shape sequences do not jump across town.
- Flood check: intersections use projected units; selected zones and water treatment are documented.
- Outlier check: inspect top and bottom 5 tracts; verify raw values and source geometry, not just final score.
- Sensitivity check: change one weight set and see whether rankings radically change. Record the result.
- Reproducibility check: teammate can rebuild or at least reload final outputs from README steps.

Known limitations to say out loud

Limitation	Plain-language explanation	Mitigation in demo
ACS uncertainty	ACS tract estimates are survey estimates and can have sizable margins of error.	Show vintage; retain MOE where possible; avoid declaring tiny differences meaningful.
Ecological fallacy	A tract average does not describe every household or parcel.	Use results for screening, then investigate at parcel/community level.
GTFS completeness	A route line is not proof of frequent, reliable, or accessible service.	State whether score is route proximity or scheduled service.
LODES timing / geography	Jobs data may lag and are block-based with disclosure protections.	State release year and aggregation; do not imply real-time openings.
FEMA scope	Flood maps are regulatory hazard references, not a complete climate-risk forecast.	Call it mapped FEMA exposure; note parcels require site-specific review.
Fairness / displacement	A need/access score can still miss displacement, accessibility, and community priorities.	Keep human review and community input as explicit next steps.

Make the map explainable

App integration

The app needs to make the data legible, not merely colorful. Keep all score logic in the data pipeline or one shared scoring module. The interface should read final GeoJSON and reveal the components behind each score.

UI element	What it should show	Why it matters
Map	Tracts colored by continuous score or transparent bands; Binghamton boundary/label; legend.	Quick geographic pattern recognition.
Click panel	GEOID/name, final score, four component scores, raw values, source years, caveat.	Makes the score inspectable.
Weights control	Preset baseline plus an alternate public-interest scenario; reset button.	Shows policy choices rather than hiding them.
Data-quality badge	Complete / partial / unavailable indicators for selected tract.	Prevents false certainty.
Method tab	Formula, definitions, sources, retrieval dates, exclusions.	Supports trust and judges' questions.
AI assistant (optional)	Only answers from current data dictionary/provenance and cites its fields.	Useful explanation layer, not an authority engine.

Safe AI integration pattern

- Provide the assistant with a compact, versioned context: data dictionary, formula, limitations, and selected tract attributes.
- Have it answer: “Why is this tract scored this way?” or “What data are missing?” Never ask it to invent parcel facts.
- Show the exact values and source-vintage labels in the UI beside generated prose.
- Use a deterministic fallback explanation template if the model is unavailable during the demo.

Suggested explanation template

“This tract's screening score is 72/100 under the baseline rubric. Housing-need priority is high (84), transit access is moderate (61), job proximity is high (80), and mapped low-flood-exposure score is 63. This is a tract-level screen; parcel conditions, zoning, and community review remain necessary.”

Show the work, not just the map

Testing and demo runbook

Functional test matrix

Test	Expected result	Fallback
Load GeoJSON	All tracts render; no console/data parse error.	Preload a static GeoJSON and a screenshot.
Click a tract	Panel matches selected GEOID and visible attributes.	Use a fixed featured-tract card.
Change weights	Score recomputes or switches scenario transparently; legend updates.	Use two precomputed scenarios.
Missing indicator	UI says unavailable / partial; no fake zero is substituted.	Show missing-data badge in static prototype.
Source link / method	Method page opens and source URLs are readable.	Keep a single local method PDF/markdown page.
Mobile / projector	Legend and tooltip remain readable.	Have browser zoom and screenshots ready.

Three-minute demo script

Time	Say / show
0:00-0:25	“Finding affordable-housing opportunities requires more than a vacant-looking lot. We built a transparent tract-level screen that combines public need, access, and risk.” Show overview map.
0:25-1:15	Click a featured tract. Show four components and raw metrics. Explain that high need is public benefit, not profit.
1:15-1:55	Toggle the baseline / alternate scenario. State that weights are visible policy choices, not hidden AI judgment.
1:55-2:35	Open methods. Show source vintages, FEMA overlay, and GTFS route-proximity caveat. Explain the one-row-per-GEOID pipeline.
2:35-3:00	Close: “CivicSite AI helps staff and community partners prioritize follow-up, then move to parcel, zoning, infrastructure, and engagement review.”

Demo rule: if data are incomplete, say so. A clear limitation increases credibility more than a smooth but unsupported claim.

Minimum final fields

Data dictionary

These names are deliberately boring and inspectable. Use snake_case; avoid one-off spreadsheet headings.

Field	Type / units	Definition	Source / transformation
geoid	string, 11 chars	Census tract identifier: state + county + tract.	TIGER/ACS key; never numeric.
tract_name	string	Human-readable ACS/TIGER tract label.	ACS NAME or boundary attribute.
acs_vintage	string	ACS 5-year release used.	Provenance.
rent_burden_35_plus_pct	number, percent	Renter households with gross rent >=35% income / selected renter-household denominator.	ACS B25070 categories; document exact sum.
poverty_pct	number, percent	Optional population below poverty threshold / appropriate population denominator.	ACS B17001-derived; document exact numerator.
transit_coverage_pct	number, percent	Tract area within declared transit buffer, or replace with service measure.	GTFS spatial overlay; record buffer.
weekday_departures_nearby	integer	Optional scheduled departures within declared stop buffer/time window.	GTFS stop_times/calendar.
workplace_jobs	integer	LODES WAC total jobs aggregated to tract or declared catchment.	LODES WAC C000 / stated vintage.
flood_area_share	number, 0-1	Selected mapped FEMA flood-zone area / tract land area.	NFHL intersection in projected CRS.
need_score	number, 0-100	Normalized housing-need component.	Formula + config.
transit_score	number, 0-100	Normalized transit component.	Formula + config.
jobs_score	number, 0-100	Normalized jobs component.	Formula + config.
low_flood_score	number, 0-100	Reverse-normalized mapped-exposure component.	Formula + config.
final_score	number, 0-100	Weighted screening score.	Formula + config.
data_status	string	complete / partial / insufficient_data.	Null and QA policy.
source_notes	string	Short, tract-level caveat / vintage pointer.	Generated from provenance.

Keep it easy to hand off

Suggested project structure

```
civicsite-ai/
README.md
data/
raw/ # untouched downloads; do not commit restricted/large files
processed/ # tract_master, final CSV, GeoJSON, GeoPackage
docs/
data_dictionary.md
provenance.csv
scoring_config.md
methodology.md
qa_report.md
notebooks/ # exploratory only
src/
ingest_acs.py
build_transit.py
build_jobs.py
build_flood.py
score_tracts.py
validate.py
qgis/
civicsite.qgz
models/ # repeatable spatial model if used
app/
public/tract_scores.geojson
src/
outputs/
map_screenshot.png
demo_fallback.pdf
```

README minimum

- Problem statement and what the score is / is not.
- Exact data sources, vintage, retrieval date, and licensing/attribution notes.
- One-command or step-by-step rebuild sequence; QGIS project instructions.
- Required dependencies and expected outputs.
- Data caveats, fairness guardrails, and test results.
- Demo URL or local run command, plus fallback screenshots.

Final deliverables checklist

- Interactive map or reliable local prototype.
- QGIS project with GTFS line layer and flood overlay.
- 3-minute demo script and one-minute backup walkthrough.
- tract_scores.geojson + tract_scores.csv.
- Data dictionary, provenance, scoring config, QA report.
- This PDF: work plan plus project guide.

Use these links

Authoritative sources and definitions

Use the freshest compatible vintage available on build day and write that choice into provenance. The URLs below are official starting points, not a substitute for recording the exact query/download you used.

Source	What to use	Official link
ACS API / variables	ACS 5-year tract estimates and variable labels. ACS variable names use table, row, and estimate/MOE suffixes.	api.census.gov/data.html ACS 5-year developer page
TIGER/Line	Census tract boundary shapefiles / geodatabases.	Census TIGER/Line files
GTFS Schedule Reference	Definitions for shapes.txt, shape_id, latitude, longitude, and sequence.	gtfs.org schedule reference
LEHD-LODES	Block-level RAC/WAC/OD job distributions, technical documentation, schema and data download.	LEHD LODES LODES code guidance
FEMA NFHL	National Flood Hazard Layer and Map Service Center mapping data.	FEMA NFHL FEMA Map Service Center

Variable-definition notes

- **ACS B25070** is the Gross Rent as a Percentage of Household Income table. Build a rent-burden percentage from documented category estimates and a matching denominator; inspect the current variable labels for your selected vintage.
- **ACS B17001** is the Poverty Status in the Past 12 Months by Sex by Age table. Its denominator must match the numerator selected; do not divide by total population unless that is the documented measure.
- **LODES WAC** associates job totals with the work census block; RAC with home block; OD with home-work pairs. Use WAC for workplace-jobs proximity unless you calculate an OD-based access metric.
- **GTFS shapes.txt** is defined by shape_id + shape_pt_sequence; points connect in increasing sequence. Longitude is X, latitude is Y, and the coordinate reference is WGS 84.
- **NFHL** is a mapped flood-hazard reference. Explain which hazard polygons were included, whether water was excluded, and whether you used tract land area or total area.

One-page release gate

Final pre-demo sign-off

- [ ] [Data] Exact source links, vintages, and accessed dates are saved.
- [ ] [Data] Every final record has one GEOID and all score components are traceable.
- [ ] [GIS] QGIS project opens with tract polygons, GTFS route lines, and flood overlay aligned.
- [ ] [Method] Formula, weights, normalization, and null policy are visible to users.
- [ ] [Ethics] UI and pitch distinguish need, suitability, public benefit, and profit; no claim of approval.
- [ ] [QA] Top/bottom tracts and extreme values were manually inspected.
- [ ] [App] Map loads locally; selected tract panel and methodology page work; fallback is ready.
- [ ] [Demo] Team has rehearsed a three-minute explanation and one limitation statement.
- [ ] [Handoff] Final GeoJSON/CSV, docs, screenshots, and this guide are in the release folder.

Definition of done: You can select any tract, explain every component in plain language, show where the number came from, and state what additional parcel-level/community review would be needed before a real decision.

CivicSite AI - Binghamton / Broome County Hackathon Guide

Prepared as a beginner-friendly, evidence-conscious build plan. Adapt the score with local stakeholders; preserve transparency as requirements change.